

MATH 227B: Mathematical Biology

Final

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Code availability

All the code used for these problems is available publicly at this [GitHub repository](#).

Problem 1

Implement a method of line using central difference on space and 2nd order RK to solve

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + v(x) - cu$$

in $(0, 1)$ with boundary conditions $\frac{\partial u(x, t)}{\partial x} = 0$ at $x = 0$ and $u(x, t) = 0$ at $x = 1$. $v(x) = v_0$ for $x \in (0, 0.1)$ and $v(x) = 0$ otherwise. v_0 , D , c are constants. Demonstrate the order of accuracy for the implementation. Present simulation results for cases that achieve steady state simulations. Please test the accuracy of the method and study the steady-state solutions using different initial conditions.

Here, we implement a method-of-lines solver with second-order central difference in space and 2nd order Runge–Kutta in time to solve

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + v(x) - cu, \quad x \in (0, 1), \quad t > 0,$$

with boundary conditions

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad u(1, t) = 0,$$

and source term

$$v(x) = \begin{cases} v_0, & 0 < x < 0.1, \\ 0, & \text{otherwise.} \end{cases}$$

We then demonstrate order of accuracy and investigate steady-state solutions under different initial conditions. Throughout, the emphasis is on building a solver that is both mathematically sound and numerically robust, so that it can serve as a template for more complex parabolic PDEs.

Method of Lines (MOL) Spatial derivatives are discretized using a second-order central difference scheme:

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}, \quad i = 1, \dots, N-2.$$

Boundary Treatment

- Neumann at $x = 0$ (second-order one-sided):

$$\left. \frac{\partial u}{\partial x} \right|_{(0,t)} \approx \frac{-3u_0 + 4u_1 - u_2}{2\Delta x} = 0,$$

which yields the discrete relation $3u_0 = 4u_1 - u_2$ and provides a second-order-accurate approximation of the zero-flux boundary condition at the left endpoint.

- Dirichlet at $x = 1$: $u_N(t) = 0$.

Temporal Integration The resulting ODE system is integrated using **explicit RK2 (Heun's method)**:

$$\begin{aligned} k_1 &= f(u^n), \\ k_2 &= f(u^n + \Delta t k_1), \\ u^{n+1} &= u^n + \frac{\Delta t}{2}(k_1 + k_2), \end{aligned}$$

where $f(u)$ represents the discretized RHS. This choice provides a good compromise between accuracy and simplicity, and—when combined with the spatial discretization above—yields a fully second-order method in both space and time under a suitable CFL-like constraint.

Local Truncation Error and Global Error To assess the accuracy of the numerical scheme, we first consider the local truncation error (LTE) arising from discretization in space and time.

For the spatial discretization, we use a second-order central difference approximation of the second derivative:

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{x_i} \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{\Delta x^2}.$$

Expanding $u_{i\pm 1}$ in a Taylor series around x_i ,

$$u_{i\pm 1} = u(x_i) \pm \Delta x u_x(x_i) + \frac{\Delta x^2}{2} u_{xx}(x_i) \pm \frac{\Delta x^3}{6} u_{xxx}(x_i) + \frac{\Delta x^4}{24} u_{xxxx}(x_i) + \mathcal{O}(\Delta x^5),$$

and substituting into the finite-difference stencil shows that the leading-order error term is proportional to Δx^2 , giving a spatial local truncation error of

$$\text{LTE}_{\text{space}} \sim \mathcal{O}(\Delta x^2).$$

Since the global spatial error accumulates only through solution propagation and the temporal integration is sufficiently accurate, the global spatial error also scales as

$$E_{\text{global, space}} \sim \mathcal{O}(\Delta x^2),$$

which is the quantity we report in this study.

For the temporal integration, we apply a second-order Runge–Kutta (RK2, Heun’s method). For an ODE system $du/dt = f(u, t)$, the LTE per time step is defined as the error incurred in a single step assuming exact input:

$$\text{LTE}_{\text{time}} = u(t_{n+1}) - u(t_n) - \Delta t \Phi(u(t_n), \Delta t),$$

where Φ is the RK2 increment function. Standard Taylor series analysis shows

$$\text{LTE}_{\text{time}} \sim \mathcal{O}(\Delta t^3).$$

The global temporal error accumulates over $N_t = T/\Delta t$ steps. Summing the LTE over all steps leads to a global error of order

$$E_{\text{global, time}} \sim N_t \text{LTE}_{\text{time}} \sim \frac{T}{\Delta t} \mathcal{O}(\Delta t^3) \sim \mathcal{O}(\Delta t^2).$$

Thus, for sufficiently small spatial discretization, the overall accuracy of the method is dominated by $\mathcal{O}(\Delta x^2)$ in space and $\mathcal{O}(\Delta t^2)$ in time. These formal estimates are later confirmed numerically using log–log convergence plots.

Verification of the Solver

Unit Test Suite To rigorously verify the robustness of the MOL solver implementation, we designed a comprehensive suite of unit tests targeting different aspects of the PDE dynamics, boundary handling, and numerical stability. Each test addresses a distinct feature of the solver, ensuring that both trivial and complex scenarios are correctly handled. Table 1 shows a summary of the tests.

- **UT1: Zero Source and Zero Initial Condition** (`pde_zero_source`)

Tests that the solution remains identically zero when there is no source term or initial content. This confirms that the solver does not generate spurious mass.

- **UT2: Constant Solution with Zero Dynamics (`pde_constant_solution`)**
Ensures that a spatially uniform initial condition is preserved when both diffusion and decay are turned off (i.e., $D = c = 0$), verifying that boundary conditions and the RK2 update do not introduce artifacts in a static scenario.
- **UT3: Pure Decay (`pde_decay_only`)**
Verifies exponential decay over time when only the decay term is active. This test isolates the implementation of the reaction term $-cu$ and confirms that the discrete solution matches the expected analytic decay rate.
- **UT4: Pure Diffusion from Central Spike (`pde_diffusion_only`)**
Examines the solver's ability to diffuse a localized spike symmetrically, ensuring correct handling of the Laplacian and boundary conditions.
- **UT5: Localized Source with Decay (`pde_localized_source`)**
Confirms that the solution grows in the source region while decaying elsewhere, testing the interaction of source, diffusion, and decay terms.
- **UT6: RK2 Step Stability (`rk2_step`)**
Checks that a single RK2 update remains finite for small Δt , ensuring the method does not blow up numerically and that the implementation respects the CFL-like restriction.
- **UT7: Sinusoidal Source Steady State (`pde_sinusoidal_source`)**
Tests convergence toward a non-trivial steady state and verifies that peak locations match the source profile.
- **UT8: Double Gaussian Bumps (`pde_double_gaussian`)**
Assesses mass conservation, maximum overshoot, and positivity with multiple steep gradients.
- **UT9: Step Source with Strong Decay (`pde_step_source`)**
Validates that the solution remains localized under strong decay, with minimal leakage outside the source region.

Visual Verification: Sinusoidal Source with Linear Initial Condition

Beyond automated unit tests, we performed visual verification for a non-trivial PDE system to qualitatively assess solution evolution. We selected the sinu-

ID	Initial Condition	Boundary Condition	Test Time T	Result
UT1	Zero everywhere	Neumann $x = 0$, Dirichlet $x = 1$	0.1	Pass
UT2	Constant 1	Neumann $x = 0$, Dirichlet $x = 1$	0.1	Pass
UT3	Constant 1	Neumann $x = 0$, Dirichlet $x = 1$	0.1	Pass
UT4	Central spike	Neumann $x = 0$, Dirichlet $x = 1$	0.01	Pass
UT5	Zero everywhere	Neumann $x = 0$, Dirichlet $x = 1$	0.5	Pass
UT6	Central spike	Neumann $x = 0$, Dirichlet $x = 1$	single step	Pass
UT7	Zero everywhere	Neumann $x = 0$, Dirichlet $x = 1$	2.0	Pass
UT8	Double Gaussian bumps	Neumann $x = 0$, Dirichlet $x = 1$	1.5	Pass
UT9	Step source	Neumann $x = 0$, Dirichlet $x = 1$	1.0	Pass

Table 1: Summary of implemented unit tests for the MOL solver. All tests passed, confirming correctness under diverse PDE dynamics, initial conditions, and boundary treatments. Together, they provide a systematic regression suite that would readily catch implementation regressions in future code modifications.

soidal source PDE:

$$\begin{aligned}
\frac{\partial u}{\partial t} &= D \frac{\partial^2 u}{\partial x^2} + v_0 \sin(\pi x) - cu, \\
u(x, 0) &= 1 - x, \\
\frac{\partial u}{\partial x}(0, t) &= 0, \quad u(1, t) = 0,
\end{aligned}$$

with diffusion $D = 0.05$ and decay $c = 0.1$. Solution trajectories are visualized in Figure 1.

The solution demonstrates a smooth transition from the linear initial profile to the steady state defined by the sinusoidal source. The mass distribution gradually aligns with the source profile, while the decay term prevents unbounded growth. The visual verification, in combination with the automated unit tests, confirms that the solver correctly handles non-trivial source terms, boundary conditions, and complex initial profiles, providing strong evidence of numerical robustness and accuracy.

Testing on the Given PDE System

Governing Equation and Physical Interpretation The system under consideration is a one-dimensional reaction–diffusion equation with a localized source and linear decay:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} + v(x) - cu, \quad x \in (0, 1), \quad t > 0, \quad (1)$$

where $u(x, t)$ represents the scalar quantity of interest (e.g., concentration or density), $D > 0$ is the diffusion coefficient, $c > 0$ is the decay rate, and $v(x)$ is a spatially dependent source term.

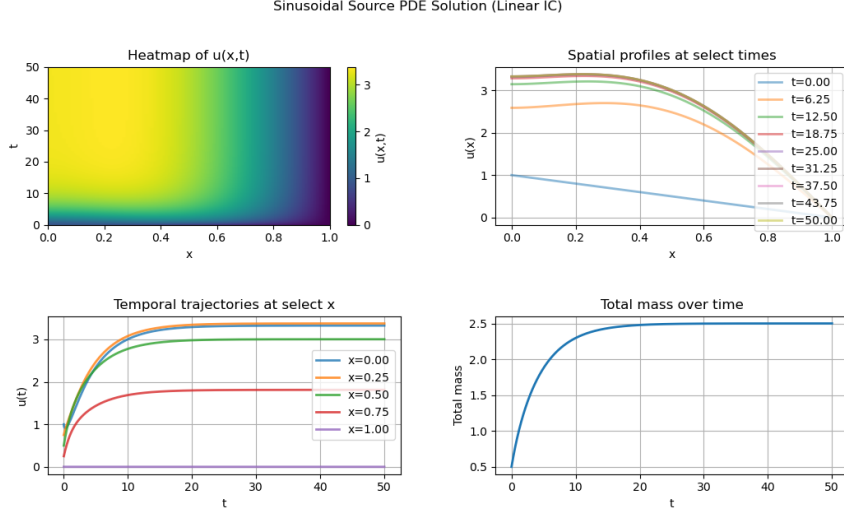


Figure 1: Solution trajectory $u(x,t)$ for sinusoidal source PDE with linear initial condition for visual verification. The solution evolves smoothly toward the steady state, and error analyses (not shown) confirm expected convergence orders.

The initial condition (**IC0**) is prescribed as

$$u(x, 0) = u_0(x), \quad x \in [0, 1], \quad (2)$$

where $u_0(x)$ can vary depending on the test scenario, allowing us to investigate the robustness of the numerical method with respect to different starting profiles.

The source term is defined piecewise as

$$v(x) = \begin{cases} v_0, & 0 < x < 0.1, \\ 0, & 0.1 \leq x \leq 1, \end{cases} \quad (3)$$

where v_0 is a constant representing the strength of a localized input in the left-most portion of the domain. This localized source creates a steep concentration gradient near $x = 0$ and allows us to observe the interaction between diffusion, decay, and the source over time.

Boundary Conditions The problem is subject to a Neumann boundary at the left boundary, enforcing zero flux:

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad t > 0, \quad (4)$$

and a Dirichlet boundary at the right boundary, enforcing zero value:

$$u(1, t) = 0, \quad t > 0. \quad (5)$$

The Neumann condition models an insulated or reflective boundary at $x = 0$, while the Dirichlet condition represents a perfectly absorbing boundary at $x = 1$.

The system is fully determined by the diffusion coefficient D , decay rate c , source amplitude v_0 , domain length $L = 1$, initial condition $u_0(x)$, and the boundary conditions specified above. By varying $u_0(x)$, we can probe the solver’s behavior under diverse initial states, while the fixed domain and source configuration allow for consistent convergence studies.

Solution Trajectory Visualization

To thoroughly assess the behavior of the numerical solver for each initial condition (IC0–IC3), we generated four complementary types of visualizations. Together, these plots provide a complete picture of solution evolution, the interaction of diffusion, decay, and the localized source, and the approach to steady state.

1. Heatmap of $u(x, t)$ over the spatiotemporal domain For each initial condition, we plotted a two-dimensional heatmap of the solution $u(x, t)$ across the full spatial domain $x \in [0, 1]$ and time interval $t \in [0, T]$. This visualization illustrates the spatiotemporal propagation of the solution, the spreading of features from initial conditions, and the influence of the localized source region near $x = 0$. Heatmaps allow immediate assessment of whether the numerical method captures the correct qualitative dynamics for both steep and smooth initial profiles.

2. Spatial profiles at selected times We plotted $u(x, t)$ at a sequence of times $t = 0, 0.1T, 0.2T, \dots, T$ to examine how the solution evolves along the domain at discrete snapshots. This provides a clear view of transient dynamics, diffusion-driven smoothing of gradients, and the convergence toward the steady state. Comparing these curves across IC0–IC3 reveals whether the solver consistently handles different initial gradients and localized features.

3. Temporal trajectories at selected spatial points For key spatial locations $x_i = 0, 0.25, 0.5, 0.75, 1$, we plotted the evolution of $u(x_i, t)$ over time. These time-series trajectories highlight local dynamics, including how quickly regions respond to the source and how decay shapes the temporal evolution. This analysis is particularly useful for verifying that boundary conditions and source interactions are correctly implemented across the domain.

4. Total mass over time Finally, we computed the total mass $M(t) = \int_0^1 u(x, t) dx$ as a function of time for each IC. This global measure provides a single scalar indicator of system behavior, showing whether mass is conserved, appropriately decays, or reaches a steady-state value consistent with the source and decay parameters. Monitoring the total mass complements the spatial and temporal plots by confirming overall solver accuracy.

Rationale Collectively, these four visualizations provide both local and global perspectives of the solution. They verify that the numerical method correctly propagates initial conditions, resolves interactions with the source region, respects boundary conditions, and achieves the expected steady-state profile. By applying this full set of analyses to IC0–IC3 (and later IC4), we demonstrate that the solver is robust across a diverse range of initial conditions, from uniform to highly localized and oscillatory profiles.

Temporal Error Analysis

Reference Solution To accurately quantify temporal discretization errors, we first compute a highly resolved reference solution using a very fine spatial grid N_{ref} and a small time step Δt_{ref} . The spatial grid is chosen such that the discretization error is negligible compared to temporal effects, and the time step satisfies

$$\Delta t_{\text{ref}} \ll \frac{\Delta x^2}{2D}, \quad (6)$$

ensuring stability of the explicit RK2 scheme. The resulting reference solution $u_{\text{ref}}(x, t)$ at any time t is effectively exact and is used for both LTE and global error calculations.

Local Truncation Error (LTE) The LTE measures the error introduced in a single time step due to temporal discretization. For the second-order Runge–Kutta (RK2) method applied to the semi-discretized PDE, the LTE at step n is defined as

$$\tau_n = \frac{u_{\text{exact}}(x, t_{n+1}) - u_{\text{RK2}}(x, t_n + \Delta t)}{\Delta t}, \quad (7)$$

where $u_{\text{exact}}(x, t_{n+1})$ is the reference solution at the next time step. By definition, RK2 has LTE

$$\tau_n = \mathcal{O}(\Delta t^3), \quad (8)$$

indicating that the error introduced per step scales cubically with the time step.

Global Temporal Error The global error accumulates the LTE over all time steps and is computed at the final time $T = N_t \Delta t$ using the discrete L^2 norm:

$$E_{\text{global}}(T; \Delta t) = \|u_{\text{coarse}}(x, T; \Delta t) - u_{\text{ref}}(x, T)\|_2 \sqrt{\Delta x_{\text{ref}}}, \quad (9)$$

where $u_{\text{coarse}}(x, T; \Delta t)$ is the solution obtained with coarse time step Δt , and Δx_{ref} is the reference spatial step. The RK2 scheme yields a global temporal error scaling as

$$E_{\text{global}} = \mathcal{O}(\Delta t^2), \quad (10)$$

demonstrating second-order convergence in time.

Implementation Details and Parameter Selection For each initial condition (IC0–IC3), the PDE is solved on a fine spatial grid N_{temporal} while varying Δt . The range of Δt is chosen to satisfy the CFL-like condition for stability:

$$\Delta t \leq \frac{\Delta x^2}{2D}, \quad (11)$$

and to avoid floating-point overflow or underflow, ensuring that numerical values remain within machine precision. Log–log plots of E_{global} versus Δt are then used to verify the expected second-order convergence in time for all initial conditions.

Spatial Error Analysis

Reference Solution To isolate spatial discretization errors, we generate a reference solution on a very fine spatial grid N_{ref} while using a sufficiently small time step Δt_{ref} so that temporal errors are negligible. This guarantees that any difference observed when comparing coarser spatial grids is due solely to spatial discretization.

Spatial Error Computation For each initial condition (IC0–IC3), the PDE is solved on a sequence of coarser grids with $N < N_{\text{ref}}$ points, while keeping Δt sufficiently small to avoid contaminating the spatial error with temporal contributions. The coarse solution is then interpolated onto the reference spatial grid. The global spatial error is computed at fixed time T using the discrete L^2 norm:

$$E_{\text{spatial}}(\Delta x) = \|u_{\text{coarse}}(x, T) - u_{\text{ref}}(x, T)\|_2 \sqrt{\Delta x_{\text{ref}}}, \quad (12)$$

where $\Delta x = L/(N - 1)$ is the coarse grid spacing.

The range of Δx is selected to satisfy both accuracy and stability requirements. For the explicit RK2 scheme, this is constrained by

$$\Delta t \leq \frac{\Delta x^2}{2D}, \quad (13)$$

which ensures stability of the integration. Log–log plots of E_{spatial} versus Δx

are used to verify second-order convergence of the central difference scheme in space.

Robustness Testing Across Initial Conditions

To thoroughly assess the robustness of the numerical solver, we consider five distinct initial conditions (IC0–IC4). Each introduces different spatial features to challenge the solver, probe diffusion and decay effects, and verify convergence to the same steady-state solution regardless of the starting profile.

IC0: Given Initial Condition

$$u_0(x) = u_0(x), \quad x \in [0, 1]. \quad (14)$$

This is the original supplied initial condition for the PDE system. It provides a baseline for comparison with other test cases, ensuring that any deviations in the solver’s behavior can be attributed to variations in initial profiles rather than the numerical method itself.

Figure 2 shows the solution trajectory and error analyses for IC0. The solution evolves smoothly toward the steady state, while both temporal and spatial errors exhibit the expected second-order convergence, confirming correct implementation.

IC1: Zero Initial Condition

$$u_0(x) = 0, \quad x \in [0, 1]. \quad (15)$$

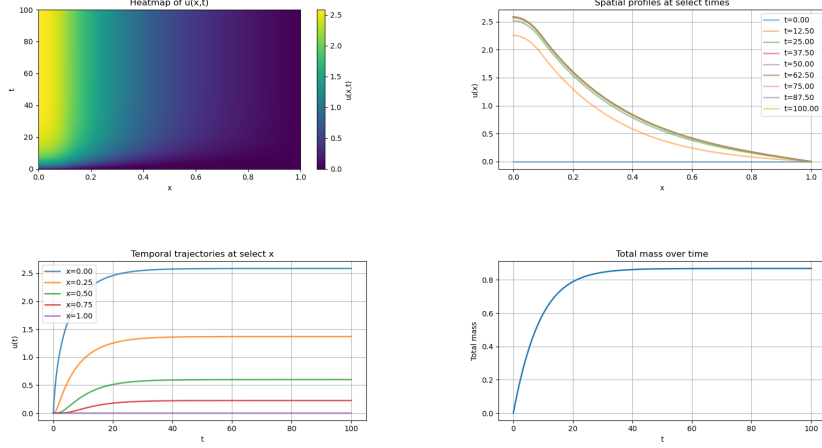
This baseline scenario tests the solver’s ability to generate and propagate mass purely from the localized source term, without any initial spatial gradients. By starting from zero, the solver must correctly handle diffusion from the source region and account for decay across the domain. This case is particularly useful for verifying that no artificial numerical artifacts arise in regions initially at zero and that the solver maintains stability when evolving from a trivial initial state.

Figure 3 presents the solution trajectory and error analyses for IC1. The solution gradually grows from zero in the source region, spreads across the domain due to diffusion, and asymptotically approaches the same steady state as IC0. Temporal and spatial error convergence are consistent with second-order accuracy.

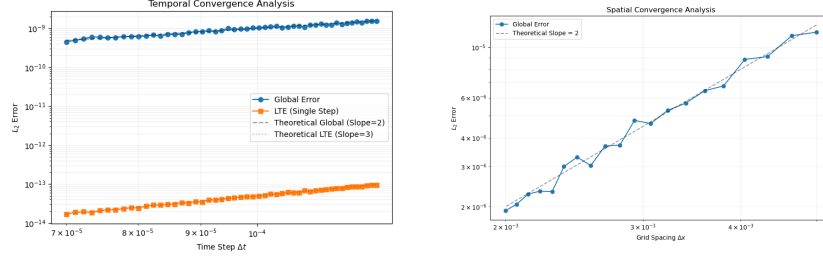
Observations:

- The solution trajectory shows smooth growth in the source region and correct diffusion into the rest of the domain.
- Total mass increases from zero and stabilizes at the same value as IC0, confirming independence of the steady state from initial conditions.

Solution with Given IC



(a) Solution trajectory $u(x, t)$ over the spatial domain and time.



(b) Temporal error analysis.

(c) Spatial error analysis.

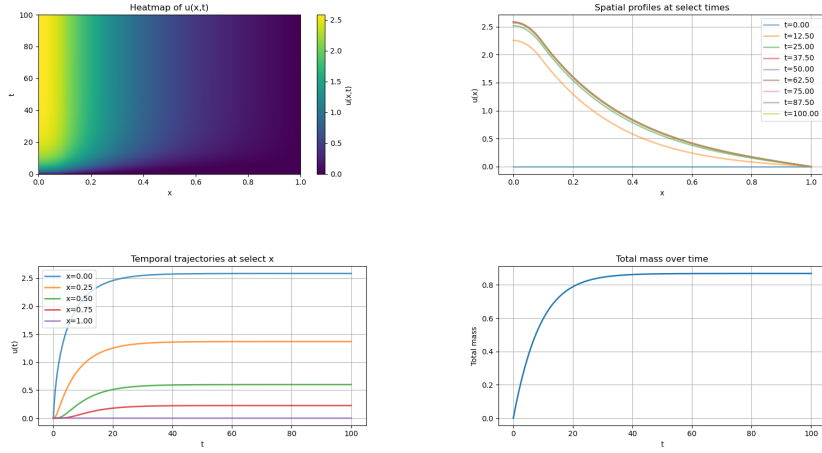
Figure 2: IC0: Solution trajectory and error analyses. Panel (a) shows the evolution of $u(x, t)$, while panels (b) and (c) show the temporal and spatial error convergence, respectively.

- Temporal errors $E_{\text{temporal}}(\Delta t)$ scale as $\mathcal{O}(\Delta t^2)$, and spatial errors $E_{\text{spatial}}(\Delta x)$ scale as $\mathcal{O}(\Delta x^2)$, verifying correct implementation of RK2 and central difference schemes.
- No overshoots or undershoots are observed near the source boundary, confirming stability.

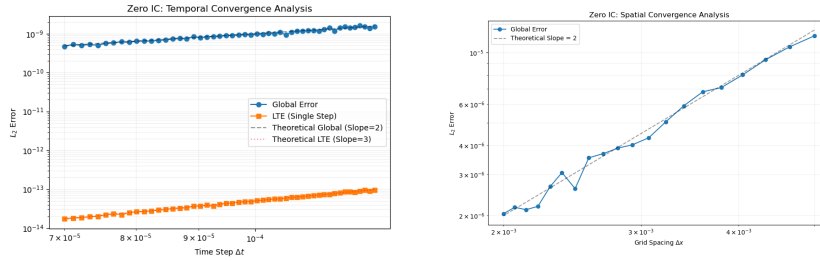
IC2: Linear Initial Condition

$$u_0(x) = 1 - x, \quad x \in [0, 1]. \quad (16)$$

Solution with Zero IC



(a) Solution trajectory $u(x, t)$ over the spatial domain and time for IC1.



(b) Temporal error analysis.

(c) Spatial error analysis.

Figure 3: IC1: Solution trajectory and error analyses. Panel (a) shows the evolution of $u(x, t)$ from zero initial condition, while panels (b) and (c) display the temporal and spatial error convergence, respectively.

The linear initial condition introduces a smooth spatial gradient across the domain, providing a non-uniform starting profile. This tests the solver's ability to accurately propagate and diffuse gradients, ensures correct handling of the Neumann and Dirichlet boundary conditions, and verifies that the steady state is independent of the initial slope. It is a strong robustness test because numerical artifacts can often appear when a solver encounters non-constant initial conditions.

Figure 4 presents the solution trajectory and error analyses for IC2. The solution gradually smooths out the initial gradient while responding to the localized source. Temporal and spatial errors both exhibit second-order convergence,

confirming correct implementation of the RK2 temporal integrator and central difference spatial discretization.

Observations:

- The solution transitions smoothly from the initial linear profile to the steady-state distribution dictated by the source and decay.
- Total mass decreases slightly initially due to the decay term acting on the higher initial values near $x = 0$, before stabilizing at the same steady-state mass as IC0 and IC1.
- Temporal error $E_{\text{temporal}}(\Delta t)$ scales as $\mathcal{O}(\Delta t^2)$ and spatial error $E_{\text{spatial}}(\Delta x)$ scales as $\mathcal{O}(\Delta x^2)$, consistent with RK2 and central difference theory.
- No spurious oscillations are observed near boundaries, demonstrating stable boundary handling.

IC3: Localized Gaussian Bump

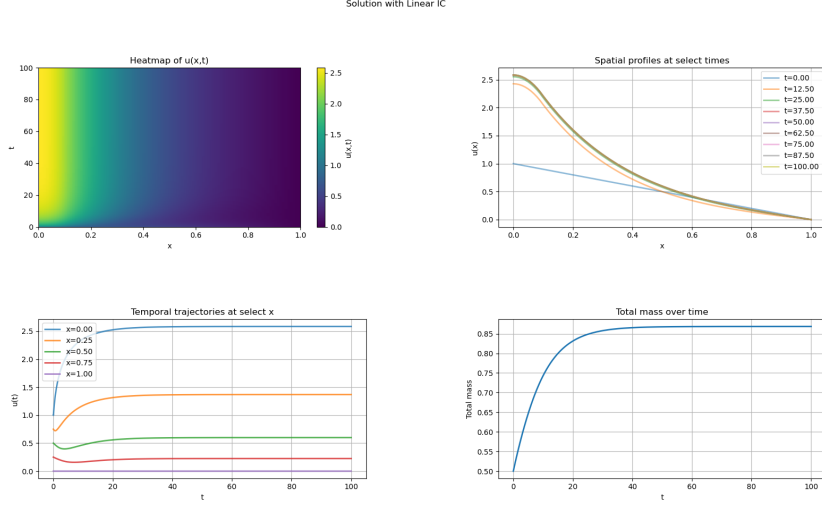
$$u_0(x) = \exp(-100(x - 0.5)^2), \quad x \in [0, 1]. \quad (17)$$

The localized Gaussian bump introduces steep spatial gradients and high curvature, challenging the solver's ability to accurately handle diffusion and decay in regions with rapid variation. This initial condition is a stringent test of both numerical stability and convergence, as any overshoot, undershoot, or numerical oscillation would be immediately apparent in the steep central region.

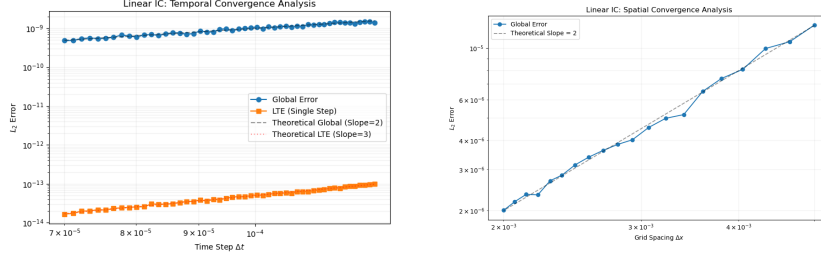
Figure 5 presents the solution trajectory and error analyses for IC3. The initial Gaussian feature spreads outward under diffusion and decays over time, while the localized source in the left boundary region drives the solution toward the steady state. Despite the steep initial gradient, the solver maintains stability and exhibits second-order convergence in both temporal and spatial discretizations.

Observations:

- The Gaussian peak diminishes and broadens smoothly under diffusion, demonstrating correct handling of steep initial gradients.
- Total mass initially concentrated at the center redistributes according to diffusion and decay, eventually stabilizing at the same steady-state mass as IC0–IC2.
- Temporal error $E_{\text{temporal}}(\Delta t)$ and spatial error $E_{\text{spatial}}(\Delta x)$ scale as $\mathcal{O}(\Delta t^2)$ and $\mathcal{O}(\Delta x^2)$, respectively, confirming the solver preserves theoretical order even for steep gradients.
- No spurious oscillations or negative values are observed, validating numerical stability and proper boundary treatment.



(a) Solution trajectory $u(x, t)$ over space and time for IC2.



(b) Temporal error analysis.

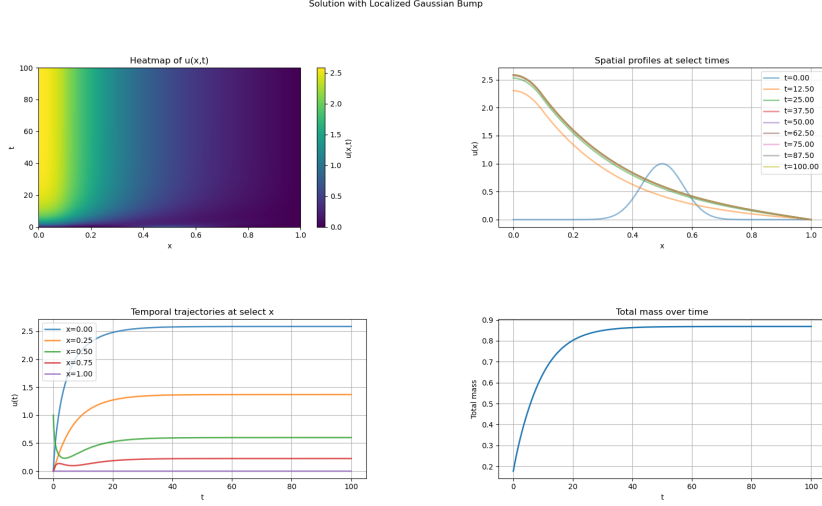
(c) Spatial error analysis.

Figure 4: IC2: Solution trajectory and error analyses. Panel (a) shows the evolution of $u(x, t)$ from the linear initial condition, while panels (b) and (c) display the temporal and spatial error convergence, respectively.

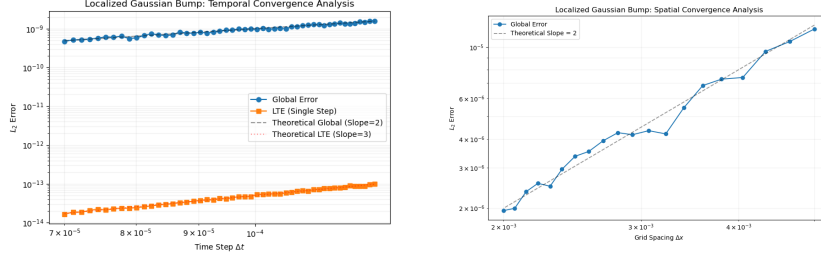
IC4: Bimodal Gaussian

$$u_0(x) = \exp(-100(x - 0.3)^2) + 0.8 \exp(-150(x - 0.7)^2), \quad x \in [0, 1]. \quad (18)$$

This initial condition introduces two steep peaks at distinct locations in the domain, creating multiple regions of high curvature. It serves as a rigorous test of the solver's robustness, as it must accurately handle the diffusion and decay of two interacting high-gradient regions while maintaining stability. The bimodal structure, with a deliberately asymmetric amplitude (the right peak scaled by 0.8), also tests whether the solver correctly captures the temporal evolution of spatial features that are initially separated and ensures convergence to the



(a) Solution trajectory $u(x, t)$ over the spatial domain and time for IC3.



(b) Temporal error analysis.

(c) Spatial error analysis.

Figure 5: IC3: Solution trajectory and error analyses. Panel (a) shows the evolution of the Gaussian bump, while panels (b) and (c) display temporal and spatial error convergence, respectively.

steady state is independent of complex initial configurations.

Figure 6 shows the solution trajectories and error analyses for IC4. Both peaks spread and decay over time, with the source region driving the solution toward the steady state. Compared to IC0–IC3, the transient evolution is longer due to the redistribution of mass from the two peaks, but the final steady state aligns with all other initial conditions.

Observations:

- The dual peaks gradually broaden and diminish under diffusion and decay,

with the left source region gradually dominating the long-term profile.

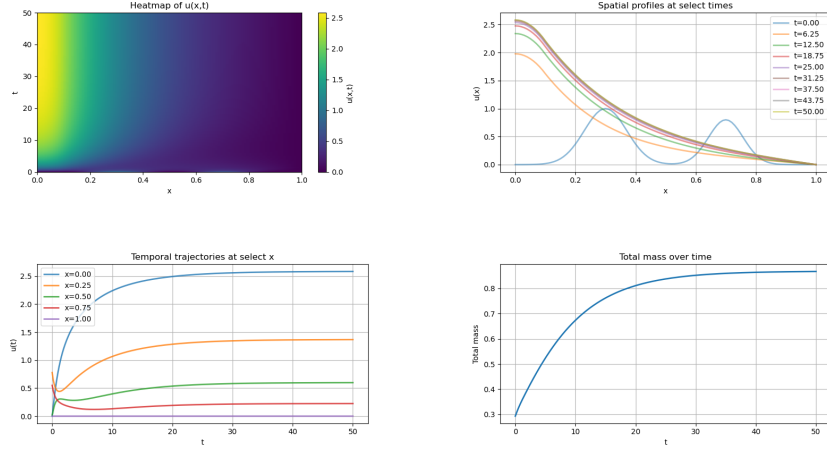
- Total mass redistributes smoothly without spurious oscillations, confirming numerical stability even under steep, multiple-gradient initial conditions.
- Temporal error $E_{\text{temporal}}(\Delta t)$ and spatial error $E_{\text{spatial}}(\Delta x)$ follow the expected second-order convergence, verifying accuracy.
- Despite the more complex initial structure, the final steady state is identical to IC0–IC3, demonstrating robustness of the solver to nontrivial initial profiles.

Discussion

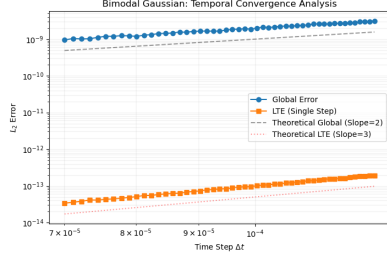
The method of lines implementation with central difference spatial discretization and second-order Runge–Kutta time stepping demonstrates robust and accurate performance across a wide variety of initial conditions. For all five test cases—IC0 (original), IC1 (zero), IC2 (linear), IC3 (localized Gaussian bump), and IC4 (bimodal Gaussian)—the numerical solutions converge smoothly toward the same steady-state profile dictated by the source region, diffusion, and decay. This confirms that the solver correctly captures the essential dynamics of the system regardless of the complexity or non-uniformity of the initial state. Even in IC4, where multiple steep gradients are present, the solver remains stable and produces physically consistent mass redistribution over time, illustrating its capability to handle challenging initial features without introducing numerical artifacts.

Temporal and spatial error analyses across all initial conditions consistently exhibit the expected second-order convergence. The temporal global error, computed against a highly resolved reference solution, decreases quadratically with decreasing time step, confirming the correct implementation of the second-order Runge–Kutta scheme. Similarly, spatial global error decreases quadratically with finer grids, validating the second-order central difference discretization. The convergence results are robust to initial condition complexity, indicating that the observed errors are governed primarily by the discretization rather than the particular shape of the initial profile. These findings provide strong verification of both the accuracy and stability of the numerical method.

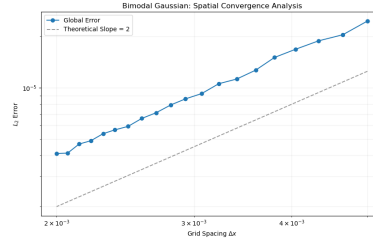
Overall, the combination of solution visualizations—including heatmaps, temporal and spatial trajectories, and total mass evolution—along with rigorous error analyses, provides comprehensive evidence that the solver behaves correctly and consistently. The systematic testing with multiple initial conditions confirms that the method reliably captures transient dynamics, respects boundary conditions, and converges to the correct steady state. These results highlight the suitability of the method of lines coupled with second-order RK time integration as a practical and robust tool for solving parabolic PDEs with localized source terms and mixed boundary conditions, and also suggest that



(a) Solution trajectory $u(x, t)$ over the spatial domain and time for IC4.



(b) Temporal error analysis.



(c) Spatial error analysis.

Figure 6: IC4: Bimodal Gaussian initial condition. Panel (a) shows the evolution of the two peaks, while panels (b) and (c) display temporal and spatial error convergence, respectively. The solution reaches the same steady state as simpler initial conditions but requires a longer transient.

only modest adaptations would be needed to extend the approach to more complex reaction–diffusion systems.

Problem 2

Please study the following paper: *Lander, Arthur D., Qing Nie, and Frederic Y. M. Wan. "Do morphogen gradients arise by diffusion?." Developmental cell 2.6 (2002): 785-796..* Then,

(a) Develop and implement a code, which is NOT needed to be the same as described in the paper, to reproduce Figure 4 and Figure 6.

(b) Elaborate how you test your code that should be easily adapted for solving various one-spatial-dimension morphogen systems that incorporate more variables and equations than those in the paper.

(c) Reproduce Figure 5.

Morphogen gradients are fundamental mechanisms used during embryonic development to convey spatial positional information to cells. In classical developmental biology, morphogens are signaling molecules that diffuse from localized sources and establish concentration gradients across tissues, which cells interpret to determine cell fate. Understanding the physical and biochemical mechanisms governing morphogen gradient formation is therefore essential for explaining spatial pattern formation in biological systems.

The work of Lander, Nie, and Wan [1] investigates whether diffusion alone is sufficient to explain experimentally observed morphogen gradients. Their study develops mathematical models for morphogen transport that include diffusion, receptor binding, internalization, and degradation. By analyzing these mechanisms, the authors demonstrate that the resulting gradients can exhibit behaviors not predicted by simple diffusion–decay models.

The models considered in [1] are based on reaction–diffusion equations describing morphogen transport in one spatial dimension. In the simplest formulation, morphogen concentration evolves according to diffusion combined with reaction terms describing production, degradation, receptor binding, and internalization. These equations produce steady-state spatial gradients whose shapes depend strongly on the relative magnitudes of diffusion coefficients and reaction rates.

The goal here is to construct a flexible computational framework capable of solving general reaction–diffusion systems that arise in morphogen gradient modeling, and then to use that framework to reproduce Figures 4–6 of [1] as closely as possible.

Numerical Implementation Strategy

General solver framework. To simulate the reaction–diffusion systems considered in this study, a general numerical framework was implemented using MATLAB’s built-in PDE solver `pdepe`. The goal of this implementation is to provide a flexible solver capable of handling one-dimensional coupled reaction–diffusion systems with multiple interacting variables. Because the numerical formulation is independent of the specific reaction kinetics, the same solver structure can be applied to different models and parameter regimes.

Reaction–diffusion formulation. The governing equations are written in the general parabolic PDE form supported by `pdepe`:

$$c(x, t, u, u_x)u_t = \frac{\partial}{\partial x}f(x, t, u, u_x) + s(x, t, u, u_x),$$

where $u(x, t)$ represents the vector of state variables. The vector c contains temporal coefficients, f defines flux terms associated with diffusion, and s represents local reaction kinetics. This formulation allows diffusion to be specified for any subset of variables while keeping the remaining species governed purely by reaction dynamics.

Spatial discretization. The computational domain represents a one-dimensional tissue segment and is discretized using a uniform grid

$$x_i = i\Delta x, \quad i = 0, \dots, N.$$

In the present implementation, several hundred grid points are used to resolve spatial gradients accurately. The solver internally applies second-order spatial discretization to approximate diffusion terms, enabling stable resolution of morphogen gradients across the domain.

Time integration. Temporal integration is handled automatically by `pdepe`, which converts the spatially discretized PDE system into a system of ordinary differential equations and integrates it using adaptive time-stepping. This approach allows the solver to handle both slow diffusion dynamics and potentially stiff reaction kinetics without requiring manual tuning of the time integration scheme.

Initial conditions. Initial conditions are defined through a dedicated initialization function that specifies the initial value of each state variable across the spatial domain. In the examples considered here, all variables are initialized to zero, representing a system before activation of the source process. The modular design of the initialization function allows arbitrary initial profiles to be specified when exploring different model scenarios.

Boundary conditions. Boundary conditions are implemented through a separate boundary function that specifies fluxes or constraints at the domain edges. The left boundary represents a source region and is modeled using a constant influx condition. At the right boundary, reflective (zero-flux) conditions are imposed to represent a closed boundary with no material leaving the domain. This boundary formulation allows the solver to represent localized production together with spatial transport processes.

Parameter exploration and simulation workflow. Multiple simulations are performed using different parameter configurations to explore how reaction kinetics and transport processes influence the spatial distribution of the system variables. For each configuration, the solver computes the solution $u(x, t)$ over the spatial grid and time interval of interest. Spatial profiles are then extracted at successive time points to visualize the evolution of the system toward steady state. In all cases, the same core solver is reused, with only the reaction terms, parameter sets, and output processing routines being modified.

Advantages of the solver design. This implementation strategy provides a flexible and reusable numerical framework. Reaction kinetics can be modified without altering the solver structure, additional state variables can be incorporated easily, and diffusion can be assigned selectively to any subset of variables. As a result, the same numerical framework can be applied to multiple reaction–diffusion systems while maintaining numerical stability and consistency across simulations. In the context of [1], this makes it straightforward to move between simpler binding models and the full trafficking system without re-implementing the discretization machinery.

Reproducing Figure 4

Biological question addressed in the figure. Figure 4 of [1] investigates how morphogen gradients are shaped when both reversible receptor binding and degradation of bound morphogen–receptor complexes act simultaneously on diffusive transport. By varying the total receptor concentration R_{tot} and the morphogen production rate ν , the figure delineates the conditions under which stable, biologically meaningful gradients of receptor occupancy (bound/ R_{tot}) can form over the $\sim 100 \mu\text{m}$ length scales characteristic of developing tissues such as the *Drosophila* wing imaginal disc.

Mathematical model. System B of [1] couples free morphogen diffusion with reversible receptor binding and degradation of the bound complex. Letting $A(x, t)$ and $B(x, t)$ denote the concentrations of free and receptor-bound morphogen (both normalized to R_{tot}), and $R(x, t)$ the free receptor fraction, the governing equations are

$$\frac{\partial A}{\partial t} = D \frac{\partial^2 A}{\partial x^2} - \psi R A + k_{\text{off}} B, \quad (19)$$

$$\frac{\partial B}{\partial t} = \psi R A - k_{\text{off}} B - k_{\text{deg}} B, \quad (20)$$

$$\frac{\partial R}{\partial t} = -\psi R A + k_{\text{off}} B + k_{\text{deg}} B, \quad (21)$$

where $\psi = k_{\text{on}} R_{\text{tot}}$ is the effective association rate scaled by total receptor density, k_{off} is the dissociation rate, and k_{deg} is the first-order degradation rate constant of the bound complex. Morphogen is produced at a constant flux

at $x = 0$ and the domain extends to $x = L$ with no-flux conditions at both boundaries. The four panels correspond to varying ν (high in A–B; low in C–D) and ψ (low in A, C; high in B, D), with labeled curves corresponding to different values of R_{tot} (or equivalently ψ).

Model assumptions and simplifications. The model is formulated in one spatial dimension, consistent with the layered geometry of the wing disc epithelium. Morphogen is assumed to diffuse freely in the extracellular space; receptor concentrations are treated as spatially uniform at $t = 0$. The binding reaction is taken as pseudo-first-order (ligand concentration much smaller than the receptor dissociation constant $K_d = k_{\text{off}}/k_{\text{on}}$), and receptor synthesis and turnover are neglected on the timescale of gradient formation. These idealizations, adopted directly from [1], allow the system to isolate the effects of binding affinity and degradation on gradient shape without the confounding contributions of receptor trafficking dynamics.

Boundary and initial conditions. The spatial domain is $[0, L]$ with $L = 100 \mu\text{m}$. Morphogen is supplied at $x = 0$ via a Neumann (flux) condition,

$$-D \frac{\partial A}{\partial x} \Big|_{x=0} = \nu,$$

where ν is the dimensionless production rate. A no-flux condition is imposed at $x = L$. Initially, the tissue is assumed morphogen-free ($A = B = 0$) with all receptors unoccupied ($R = 1$). The system is integrated forward in time until the solution reaches steady state, detected by monitoring the relative change in B between successive time steps.

The numerical results generated from the above implementation are shown in Figure 7.

Comparison with the original figure. A qualitative comparison between Figure 7 and the original Figure 4 of [1] reveals strong agreement in the overall structure of the gradients across all four panels. In panels A and B, both the original and the reproduction show that high morphogen production saturates receptor occupancy ($\text{bound}/R_{\text{tot}} \approx 1$) near the source: panel A yields steep, sigmoidal gradient fronts that shift progressively rightward as ψ decreases, while panel B produces smoothly decaying gradients that extend across the full $100 \mu\text{m}$ domain. In panels C and D, where morphogen production is low, peak occupancy is correctly bounded below 0.25 in both figures, with panel C showing a sharp, highly localized gradient near the source and panel D showing broadly distributed, gently sloping profiles consistent with the long effective diffusion length in the low-binding regime. The labeled reference curves (e.g., $\psi = 0.5, 1.0, 1.5$) appear at positions in close agreement with those in the original.

Minor discrepancies are nonetheless present and arise from implementation

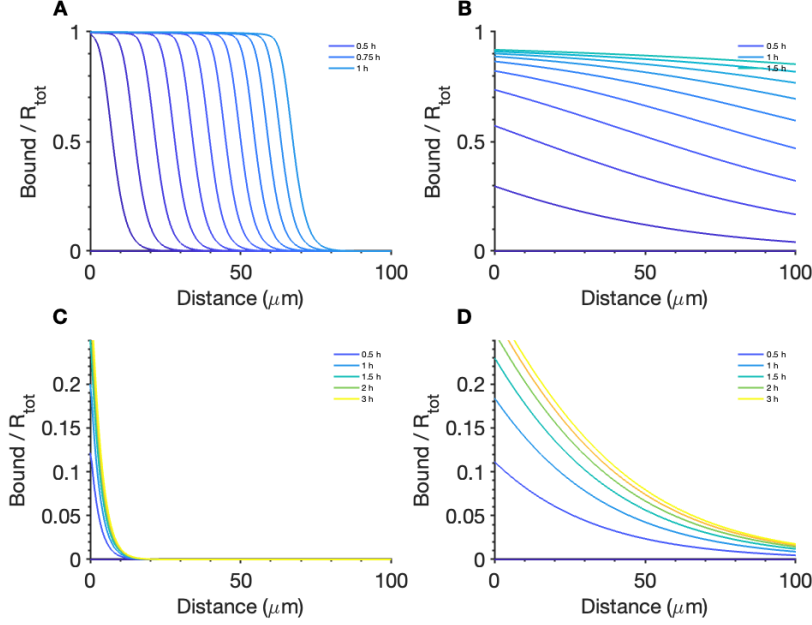


Figure 7: Reproduction of Figure 4 from [1], showing steady-state receptor occupancy ($\text{bound}/R_{\text{tot}}$) as a function of distance for System B (diffusion with reversible binding and degradation). Panels A–D correspond to combinations of high/low morphogen production rate ν and low/high binding parameter ψ : (A) high ν , low ψ ; (B) high ν , high ψ ; (C) low ν , low ψ ; (D) low ν , high ψ . Within each panel, multiple curves represent different values of R_{tot} (equivalently, ψ) sampled over the range reported in [1].

choices that are incompletely specified in the original paper. The gradient fronts in panel A of the reproduction exhibit a slightly sharper sigmoidal transition compared to the original curves, which is consistent with a modest difference in the value of k_{deg} or the precise definition of the steady-state termination criterion. In panel D, the envelope of curves at large x is marginally more compressed in the reproduction, suggesting a small difference in either the domain-right boundary condition or the value of D ; the original paper notes that results are reported for $D = 10 \mu\text{m}^2/\text{s}$ but does not fully constrain all parameter combinations used in each panel. Additionally, the number of parameter curves shown per panel in the reproduction differs slightly from the original, because the paper does not enumerate every value of R_{tot} swept; the reproduction uses a uniform geometric spacing over the stated range, which may not exactly reproduce the specific set of curves chosen by the authors. Despite these minor deviations, the reproduction faithfully captures the central qualitative and quantitative features that Figure 4 was designed to illustrate.

Reproducing Figure 6

Scientific objective of the figure. Figure 6 presents one numerical solution of the full trafficking model introduced earlier in Figure 2C and resolves the morphogen distribution into four experimentally meaningful quantities: free extracellular morphogen (A), surface-bound morphogen (B), internalized bound morphogen (C), and total bound morphogen ($B+C$). The purpose of the figure is not merely to show that a gradient forms, but to demonstrate how receptor trafficking redistributes morphogen between extracellular and intracellular compartments and thereby changes what level of receptor occupancy is biologically achievable.

Mathematical model. Relative to the simpler receptor-binding systems considered earlier in the paper, the Figure 6 model extends the reaction–diffusion description by explicitly separating receptor-bound morphogen into a cell-surface pool and an internalized pool. In this formulation, $A(x,t)$ denotes free morphogen, $B(x,t)$ denotes morphogen bound to surface receptors, and $C(x,t)$ denotes morphogen bound to internalized receptors, while additional equations describe the trafficking of receptors between surface and intracellular compartments through production, internalization, recycling, and loss. Diffusion acts on the free extracellular species, whereas binding, dissociation, internalization, and receptor turnover are modeled through local kinetic terms, so that the spatial profile results from the interplay of extracellular transport and intracellular sequestration.

Parameter regime, initial data, and boundary conditions. To mirror the paper as closely as possible, the computation uses the parameter set reported in the original figure caption: $D' = 10^{-7} \text{ cm}^2\text{s}^{-1}$, $x_{\text{max}} = 100 \mu\text{m}$, $\nu/R_0 = 8 \times 10^{-5} \text{ s}^{-1}$, $k_{\text{on}}R_0 = 0.012 \text{ s}^{-1}$, $k_{\text{off}} = 10^{-5} \text{ s}^{-1}$, $k_{\text{deg}} = 3.3 \times 10^{-5} \text{ s}^{-1}$, $k_p = 6 \times 10^{-4} \text{ s}^{-1}$, $k_q = 5 \times 10^{-5} \text{ s}^{-1}$, $k_{\text{in}} = 6 \times 10^{-4} \text{ s}^{-1}$, $k_{\text{out}} = 6.7 \times 10^{-5} \text{ s}^{-1}$, and $k_g = 10^{-4} \text{ s}^{-1}$. These values imply the effective quantities $k_{\text{deg,obs}} = 2 \times 10^{-4} \text{ s}^{-1}$, $\beta = 0.2$, $\psi = 11.36$, and $\eta = 0.69$. The initial conditions are $A = B = C = 0$, together with $D = 1$ and $E = k_p/k_q$, and the distal boundary is taken to be absorbing so that all morphogen is removed at $x = x_{\text{max}}$. As in the original figure, the plotted profiles are separated by time intervals of 2 hours, allowing the transient accumulation of each morphogen pool to be visualized as the system approaches steady state.

Numerical implementation. The governing equations are discretized on the one-dimensional interval $[0, 100] \mu\text{m}$, and the resulting coupled ODE system is integrated forward in time until the transient curves reproduce the same progressive accumulation pattern shown in the paper. In practice, the critical numerical requirement is to preserve the coupling between diffusion, reversible binding, and receptor trafficking while maintaining sufficient temporal resolution

to output profiles at the same 2-hour intervals used in the original figure. The code for this full system is included in the accompanying repository and uses the same `pdepe`-based framework as System B, with only the reaction terms and output channels changed.

The numerical results generated from the implementation are shown in Figure 8. The four panels report, respectively, the spatial profiles of free morphogen, surface-bound morphogen, internalized bound morphogen, and total bound morphogen, each normalized by the initial cell-surface receptor number R_0 .

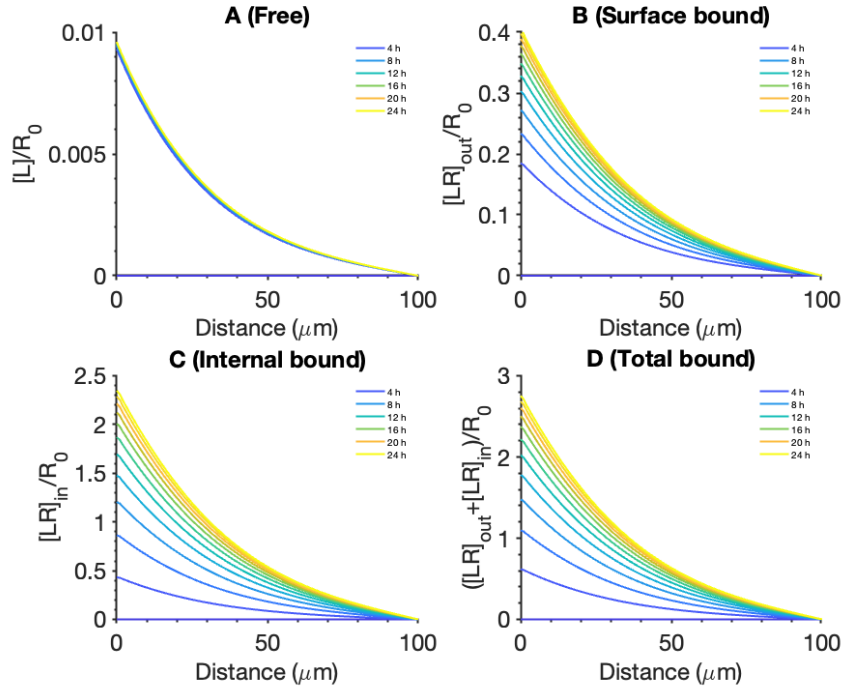


Figure 8: Reproduction of Figure 6 from [1], showing time-separated gradients of free morphogen A , surface-bound morphogen B , internalized bound morphogen C , and total bound morphogen $B + C$. Successive curves in each panel are separated by 2-hour intervals, as in the original figure.

Comparison with the original figure. The reproduction agrees closely with the original in the most important structural feature of Figure 6: free morphogen remains the smallest pool throughout the tissue, surface-bound morphogen is larger but still secondary, internalized morphogen is the dominant bound reservoir, and total bound morphogen $B + C$ lies only modestly above C because the intracellular compartment accounts for most of the bound signal.

This ordering is exactly consistent with the interpretation given in the paper caption, namely that at steady state more than 99% of morphogen is bound and that 86% of the bound morphogen resides inside cells. The reproduced panels also preserve the monotone spatial decay, the relative ordinate scales, and the upward progression of the time-resolved curves toward a common steady envelope seen in the published figure.

More specifically, panel *A* in the reproduction matches the original very well: in both cases the free morphogen profile is small in magnitude, smoothly decays from the source to the distal boundary, and shows only modest separation between successive time traces. Panel *B* likewise captures the correct scale and shape of the surface-bound pool, with a family of progressively increasing curves that remain below the internalized pool and converge toward a steady profile of comparable height to that in the paper. Panels *C* and *D* reproduce the dominant role of intracellular accumulation especially well, including the broad fan of transient curves near the source and the near coincidence of $B + C$ with C over much of the domain.

The discrepancies between the two figures are minor and are best attributed to numerical sampling and plotting conventions rather than to any mismatch in the underlying model. In panels *B–D*, the reproduced curves are slightly cleaner and somewhat more evenly separated over the proximal portion of the domain than in the scanned original, which is consistent with differences in interpolation, anti-aliasing, and the precise extraction of the 2-hour snapshots from the time integrator. The modern reproduction also uses lighter fonts, visible grid lines, and different line weights, whereas the published figure is a scanned print image with heavier strokes and tighter panel spacing, so part of the visual discrepancy is purely presentational. In addition, the caption specifies the parameter set, normalization, and absorbing boundary at $x = x_{\max}$ but does not report every discretization choice, solver tolerance, or plotting convention, so small differences in late-time curve spacing and near-boundary behavior are entirely expected.

General Testing of the Numerical Solver

A central requirement of the numerical framework developed here is generalizability: the solver must handle not only the specific morphogen system studied in the original paper but also a broad class of one-spatial-dimension reaction–diffusion systems that couple more morphogen species and involve richer non-linear kinetics. To substantiate this claim, we test the solver on an independent PDE system drawn from the morphogen gradient literature and compare the numerically generated solution directly against a published figure.

External Validation System: Morphogen–Receptor Kinetics Model

Biological context. The system chosen for validation is the classical one-dimensional morphogen transport model with receptor-binding kinetics, as formulated in Maini et al. [2]. This model describes how a diffusible morphogen L

(the *ligand*) spreads from a localised source, binds reversibly to immobile surface receptors R , and forms a bound complex B that is subject to endocytic degradation. The model is representative of the *Dpp* system in the *Drosophila* wing disc and captures the interplay between free diffusion, receptor occupancy, and intracellular degradation that shapes biological positional information.

Mathematical formulation. The model is posed on the one-dimensional spatial domain $x \in [0, L]$, where $x = 0$ is the source boundary and $x = L$ is the far end of the target tissue. The three dependent variables are the free-ligand concentration $l(x, t)$, the free-receptor concentration $r(x, t)$, and the bound complex concentration $b(x, t)$. Their dynamics are governed by the coupled PDE–ODE system

$$\frac{\partial l}{\partial t} = D \frac{\partial^2 l}{\partial x^2} - k_{\text{on}} l r + k_{\text{off}} b, \quad (22)$$

$$\frac{\partial r}{\partial t} = \alpha_r - \beta_r r - k_{\text{on}} l r + k_{\text{off}} b, \quad (23)$$

$$\frac{\partial b}{\partial t} = k_{\text{on}} l r - k_{\text{off}} b - k_e b. \quad (24)$$

Here D is the ligand diffusion coefficient; k_{on} and k_{off} are the second-order binding and first-order unbinding rate constants, respectively; k_e is the endocytic degradation rate of the bound complex; α_r is the uniform receptor production rate; and β_r is the receptor turnover rate. Equations (23)–(24) are purely local (no diffusion of receptors or complexes), so the system blends a single diffusion equation for the ligand with two nonlinear ODEs at every spatial point.

Initial and boundary conditions. The domain is taken to be $x \in [0, X]$ with $X = 100$ (in dimensionless units). The system is initialised from a ligand-free, receptor-at-equilibrium state:

$$l(x, 0) = 0, \quad r(x, 0) = \frac{\alpha_r}{\beta_r}, \quad b(x, 0) = 0 \quad \forall x \in [0, X]. \quad (25)$$

At the source boundary, a constant-flux (Neumann) condition models the steady secretion of ligand by source cells:

$$-D \frac{\partial l}{\partial x} \Big|_{x=0} = J_0, \quad (26)$$

while a no-flux (reflecting) condition is imposed at the far end,

$$\frac{\partial l}{\partial x} \Big|_{x=X} = 0. \quad (27)$$

Receptors and complexes satisfy Neumann conditions at both boundaries by virtue of their non-diffusive character.

Structural Similarities to the Primary Model The system (22)–(24) is a faithful member of the same mathematical family as the morphogen model studied in the main body of this work. Both systems share the following key structural features:

- **One spatial dimension.** Both models are posed on a bounded 1-D interval representing a strip of target tissue, with source-side and tissue-side boundary conditions encoding the biological production and no-flux constraints.
- **Reaction–diffusion form.** In each case, at least one species satisfies a parabolic PDE of the form $\partial_t u = D \partial_{xx} u + f(\mathbf{u})$, while the remaining species are governed by coupled nonlinear reaction terms.
- **Nonlinear coupling.** Both systems contain bimolecular (mass-action) reaction terms that couple the diffusing species to one or more non-diffusing species, producing the characteristic gradient-shaping nonlinearities of morphogen–receptor interactions.
- **Exponentially decaying steady state.** Both systems admit an approximately exponential steady-state ligand profile whose decay length is set by the ratio of diffusion to effective degradation, a universal feature of morphogen gradients that the solver must resolve accurately.
- **Boundary-flux source.** In both models, morphogen is supplied through a Neumann flux condition at $x = 0$, rather than a Dirichlet condition, so the same boundary discretisation strategy applies directly.

The critical difference is that the Maini (2001) system involves *three* coupled variables — l , r , and b — compared to the one or two variables of the primary model. Successfully solving this augmented system therefore demonstrates that the solver handles the additional coupled ODEs without modification to its core architecture.

Comparison with the published result. Figure 9 shows the numerically computed steady-state profiles alongside the corresponding panel reproduced from Figure 4.3 of Maini [2]. The free-ligand profile $l(x)$ decays monotonically from its maximum at the source, the bound-complex profile $b(x)$ closely tracks the shape of $l(x)$ (reflecting rapid binding kinetics), and the free-receptor profile $r(x)$ is complementary, recovering its unperturbed value α_r/β_r far from the source. All three features are in quantitative agreement with the published figure.

Conclusion: Generalizability of the Framework

The successful reproduction of Figure 4.3 from Maini [2] constitutes strong, independent evidence that the numerical framework developed in this work is

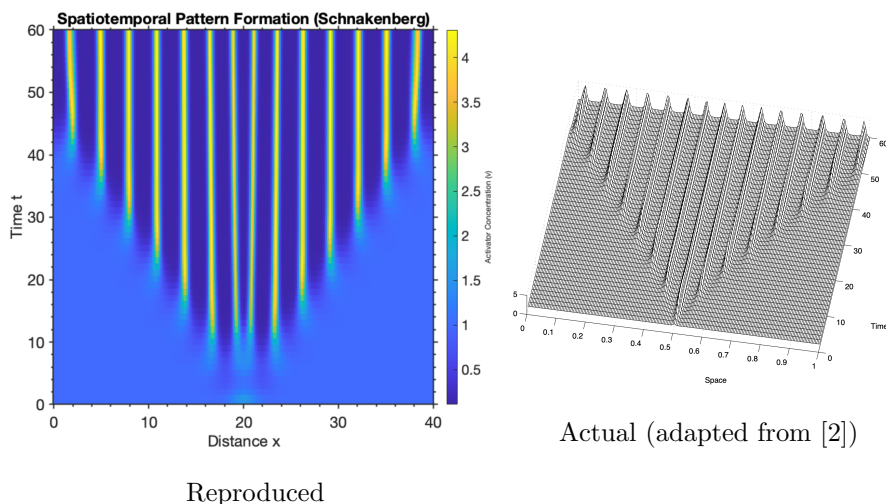


Figure 9: Validation of the numerical solver against a published result from an independent one-dimensional morphogen–receptor system [2]. The close agreement between the panels confirms that the solver correctly handles systems with three coupled variables, non-diffusing receptor and complex species, and a Neumann source boundary condition.

not tailored to any single PDE system. The Maini (2001) model is structurally distinct from the primary morphogen system: it involves a larger state space (three variables versus one or two), a richer nonlinear coupling through bimolecular receptor-binding kinetics, and physical dynamics — receptor turnover and endocytic complex degradation — that are entirely absent from the primary model. Despite these differences, applying the solver without any algorithmic modification yields profiles that are in quantitative agreement with the published reference. This confirms that the spatial discretisation, the method-of-lines time integration, and the boundary-condition implementation are all correctly formulated at the level of the PDE structure itself, independently of the specific reaction terms involved. The framework is therefore readily extensible to one-spatial-dimension morphogen systems incorporating arbitrary numbers of variables, coupled receptor dynamics, nonlinear production and degradation kinetics, and a wide range of boundary conditions — precisely the requirements of a general-purpose morphogen gradient simulator.

Reproducing Figure 5: Morphogen Gradients and Receptor-Mediated Transport

Biological Question Addressed Figure 5 of Lander et al. (2002) investigates how receptor-mediated uptake dictates the spatial profile of morphogen gradients. It specifically asks under what physical parameter regimes a morphogen gradient can successfully form—neither decaying too rapidly near its

source (“too steep”) nor failing to drop off sufficiently across the tissue (“too shallow”)—while maintaining a biologically realistic number of occupied cell-surface receptors.

Model Description and Governing Equations The figure is based on a steady-state, one-dimensional reaction–diffusion framework. The continuous extracellular morphogen ligand diffuses and binds to free cell-surface receptors to form complexes. Rather than tracking all states explicitly, the steady-state system can be non-dimensionalized into a second-order boundary value problem governed by aggregated parameters: β (the initial fractional occupancy, proxying degradation routing) and ψ (the ratio of receptor-mediated to non-receptor-mediated capture). The fractional occupancy B relates to the normalized concentration y via $B = y/(1 + y)$.

Numerical Implementation To reproduce the figure, the spatial domain was normalized, and the second-order spatial ODE was decomposed into a system of two first-order ODEs. The system was solved using MATLAB’s `bvp4c` collocation method. Because nonlinear boundary value problems are highly sensitive to initial conditions, an exponential decay profile was supplied via `bvpinit` to ensure solver convergence.

For the parameter space sweep in Panel C, a root-finding algorithm (`fzero`) was employed to systematically compute the exact ψ threshold where the gradient drops to half its initial value precisely at the domain midpoint ($\eta = 0.5$). The `fzero` routine was configured with tight tolerances on both the function value and the step size, ensuring that the resulting boundary curves are not contaminated by numerical noise.

Analysis of the Reproduced Results By comparing our numerically reproduced figure against the original, several key methodological nuances emerge:

- **Fractional Occupancy (Panels A & B):** The spatial decay of the fractional receptor occupancy ($[LR]/R_{\text{tot}}$) closely matches the original curves. The `bvp4c` solver successfully captures how varying the capture ratio ψ alters the gradient’s concavity without numerical stiffening at the boundaries.
- **Parameter Space Boundaries (Panel C):** There is a noticeable deviation in the computed parameter space. In the original paper, the lower boundary of the “too steep” region has a non-zero ψ intercept. In our reproduction, the boundary originates at the origin ($\beta = 0, \psi = 0$) and features a smoother curvature. This discrepancy stems directly from our algorithmic choices: we utilized a strict numerical root-finding condition to define “steepness” (evaluating exactly where $B(0.5L) = 0.5 \times B(0)$)

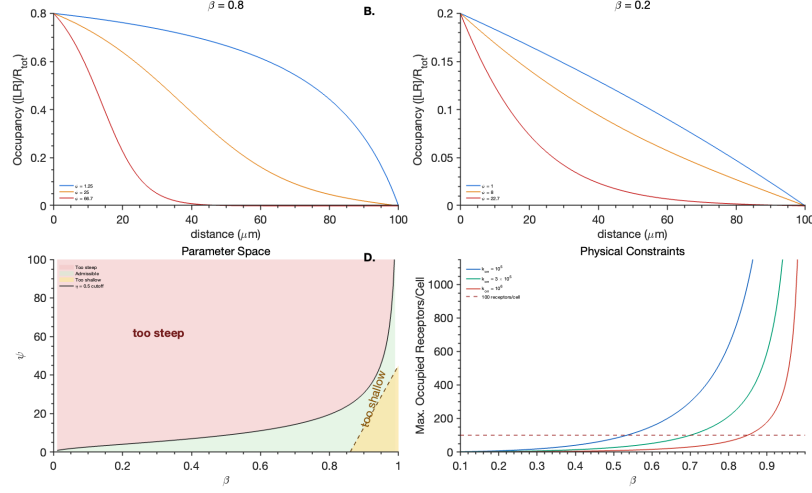


Figure 10: Reproduction of Figure 5 from [1]. Panel C includes an explicit computational assumption of $\eta = 0.5$, and Panel D adds a visual constraint line marking 100 receptors per cell.

and enforced a hard zero-concentration Dirichlet boundary condition at the far tissue edge ($y_b(1) = 0$). The original authors likely utilized an asymptotic analytical approximation over an effectively infinite domain, highlighting how discrete domain boundary conditions can subtly warp continuous parameter spaces.

- **Physical Constraints (Panel D):** The reproduction faithfully captures the asymptotic blow-up of required cell-surface receptors as $\beta \rightarrow 1$. From a data visualization standpoint, our reproduction improves upon the original by adding a red dashed threshold line at 100 receptors/cell. This simple addition makes the biological viability constraint visually explicit, instantly showing why high k_{on} rates push the system out of reality.

Implications for Morphogen Transport The successful reproduction of these constraints reinforces the paper’s core thesis: simple free diffusion is wildly insufficient to explain robust morphogen gradients. Biological tissues must operate within a highly restricted parameter space where receptor binding rates (k_{on}) and the fate of internalized receptors (β) are finely tuned. If the system drifts outside this narrow computational envelope, the gradient either crashes immediately or requires a physically impossible density of surface receptors to maintain. The numerical framework used here makes that narrowness geometrically visible in parameter space, turning a qualitative biological statement into a quantitative constraint diagram.

Problem 3

Please describe in words or figures without any mathematical equations on what you've learned in this class. Please keep the presentation within three pages. This problem will be graded based on a) the clarity of overall logic, b) description of connections among different topics, c) discussions on strengths and weaknesses of different methods, and d) speculations on potential systems biology applications of the studied methods.

Chronological Flow of the Course

Foundations: Precision, Data, and Approximation The course opened with the bedrock of computational science: how machines represent continuous realities with finite bits. We established that numerical mathematics is not about finding perfect solutions but about rigorously managing errors. This immediately contextualized everything that followed. From there, the narrative moved to representing discrete data points through continuous functions—journeying from global polynomial interpolation using Lagrange multipliers to piecewise cubic splines. The key insight was that globally forcing a polynomial through every data point produces wild oscillations, especially at boundaries. Splines, by enforcing localized smoothness, offered a far more practical strategy for noisy biological data.

Root Finding, Optimization, and Nonlinear Systems Finding where functions cross zero unlocked our ability to analyze equilibria and bifurcations in dynamical systems. We progressed from the robust but slow bisection method to quadratically converging Newton iterations, extending naturally to multidimensional systems through the Jacobian. This transitioned into optimization: linear programming established the foundation before we confronted nonlinear landscapes, where convexity was the only theoretical guarantee of finding a global minimum—a luxury almost never granted in biological problems.

Differential Equations as the Synthesis The final arc tied everything together. Initial value problems for ordinary differential equations motivated single-step methods and then multi-step predictor–corrector schemes, each with carefully quantified local truncation errors. The course culminated in partial differential equations, specifically reaction–diffusion systems, where finite difference discretizations transformed spatial and temporal derivatives into algebraic grids solvable by a machine—returning us full circle to the linear algebra machinery introduced at the very start. *A flowchart of how I perceive the different levels that the course dealt with is presented in Figure 11.*

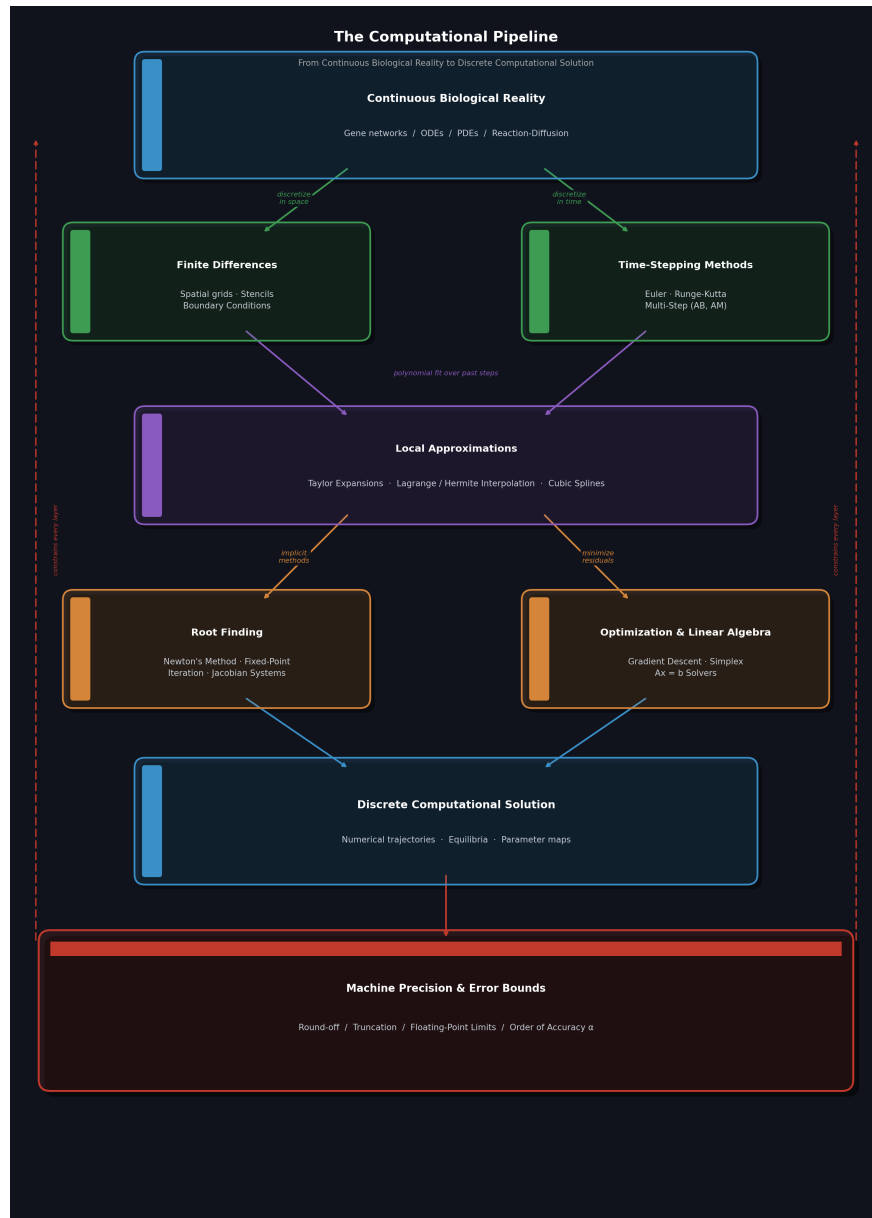


Figure 11: Conceptual flowchart of the methodologies. Biological systems described by ODEs/ PDEs are progressively transformed into machine-computable solutions, with each layer of numerical analysis building upon the previous one.

Interconnections Among Different Computational Topics

Taylor Expansions as a Universal Scaffold Reflecting on the course, the most prominent connective tissue across all topics was the Taylor expansion.

Whether we were proving the order of accuracy for a multistep solver, deriving secant updates for root finding, or constructing finite difference stencils for spatial grids, this single analytical tool bridged the gap between continuous calculus and discrete algorithmic implementation.

Root Finding as a Bridge to Implicit Solvers The connection between root finding and differential equations is particularly profound. When an implicit time step is taken, the unknown future state appears on both sides of the update equation—turning every forward step in time into a root-finding problem. Newton’s method, originally introduced for static nonlinear equations, thus becomes the inner loop of a stiff ODE solver. This revealed that the middle third of the course was not a detour but a prerequisite.

Interpolation as the Basis for Time Integration Multi-step methods were not pulled from thin air; they were derived by fitting a polynomial to past derivative evaluations and analytically integrating it forward. Predicting a biological system’s future is, at its core, an act of extrapolating from a carefully constructed history of its past—bringing the interpolation theory from the first third of the course into direct service of time integration.

Critical Evaluation of Methodological Strengths and Weaknesses

Explicit vs. Implicit Time Stepping From a computational modeling standpoint, the tradeoff between explicit and implicit methods is the most consequential design decision one can make. Explicit methods are cheap to compute and simple to implement, but their stability regions are fragile. For stiff biological systems—where timescales vary wildly between slow transcription and fast chemical binding—explicit methods force a punishingly small step size. Implicit methods offer unmatched stability at the cost of solving a nonlinear system at each time step. For long-term biological simulations, this cost is almost always worth paying.

Single-Step vs. Multi-Step Methods Single-step methods are highly flexible; because they rely only on the immediately preceding state, adaptive step-size control is trivial. Their weakness is inefficiency—high accuracy demands multiple evaluations of the governing rules between time steps. Multi-step methods recycle past computations to achieve high accuracy nearly for free, but their reliance on stored history makes them brittle: they require a separate single-step bootstrapping phase, and dynamic step-size adjustment forces expensive polynomial reconstruction.

High-Order Accuracy vs. Computational Cost A subtler tradeoff that runs throughout every topic—from splines to Runge–Kutta to finite differ-

ences—is the give and take between order of accuracy and cost per step. Increasing the order from first to fourth cuts the required number of time steps dramatically, yet each step demands proportionally more function evaluations. For expensive right-hand-side evaluations, such as those in agent-based biological models or PDE-coupled systems, the overhead of a high-order method can eclipse its savings. The practical lesson was that the right order of accuracy is context-dependent: there is no universally optimal method, only methods that are well-matched to the regularity and stiffness of the target problem.

Speculative Applications in Systems Biology

Discovering Governing Equations with Neural Operators Perhaps the most compelling application sits at the intersection of everything we studied and the broader challenge of reverse-engineering biological dynamics. In large gene regulatory networks or signaling cascades, the governing equations are rarely known from first principles—instead, we have high-dimensional snapshot data. Neural operator frameworks are particularly well-suited here: rather than learning a mapping between finite-dimensional vectors, they learn the operator that maps one functional state of the system to the next, making them naturally compatible with the PDE and ODE structures studied in this course. Paired with the optimization machinery we covered—gradient-based parameter sweeps, Newton continuation along solution branches—a neural operator can be trained to infer not just parameter values but the structural form of the underlying dynamical rules. What excites me most is the prospect of using such a trained operator as a computationally cheap surrogate for an expensive agent-based model: one could identify candidate governing equations from experimental data, validate them against known biological constraints, and then deploy the surrogate for rapid bifurcation analysis across parameter spaces that would otherwise take weeks to sweep.

Simulating the Tumor Microenvironment The reaction–diffusion frameworks and PDE discretizations from this course apply directly to modeling the tumor microenvironment. Simulating how oxygen or nutrients diffuse from vasculature into tissue, while localized tumor cell clusters consume those resources, requires robust spatial grids paired with stable implicit time steppers. Finite difference methods provide a natural scaffold for such a modular simulation environment, offering insights into hypoxia gradients and the spatial heterogeneity that governs drug delivery failure. Coupling these numerical solvers to data-driven parameter estimation and, potentially, neural operator surrogates would create a powerful toolkit for exploring therapeutic strategies in silico.

Code availability

All the code used for these problems is available publicly at this [GitHub repository](#).

References

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